

Conditional Probability

GCSE Maths · Worked Solutions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

A bag contains 5 red and 3 blue counters. One red counter is taken out and not replaced.
How many counters are left in the bag?

1 There were $5 + 3 = 8$ counters.

2 $8 - 1 = 7$

Answer: 7

2 [2 marks]

A bag contains 5 red and 3 blue counters. One red counter is taken out and not replaced.
Work out the probability that the next counter taken is also red.

1 After removing a red, there are 4 reds left.

2 The total is now 7 counters.

Answer: $\frac{4}{7}$

3 [2 marks]

Ten cards numbered 1 to 10 are shuffled. One card is taken and **not** replaced; it is the 3.
Work out the probability that the next card taken is even.

1 The even cards are 2, 4, 6, 8 and 10, so 5 of them — and the 3 was odd, so none has gone.

2 There are now 9 cards left.

Answer: $\frac{5}{9}$

4

[2 marks]

The table shows information about 40 students.

	Likes science	Does not like science	Total
Likes maths	15	9	24
Does not like maths	6	10	16

A student who likes maths is chosen at random. Work out the probability that they also like science.

1 The condition restricts us to the 24 students who like maths.

2 Of those, 15 also like science.

3 $\frac{15}{24} = \frac{5}{8}$

Answer: $\frac{5}{8}$ (0.625)

5

[3 marks]

A bag contains 4 green and 5 yellow counters. Two are taken without replacement.

Work out the probability that both are green.

1 $P(\text{first green}) = \frac{4}{9}$

2 After a green is removed there are 3 greens out of 8 counters: $\frac{3}{8}$

3 Multiply: $\frac{4}{9} \times \frac{3}{8} = \frac{12}{72}$

4 $= \frac{1}{6}$

Answer: $\frac{1}{6}$

6

[3 marks]

A bag contains 4 green and 5 yellow counters. Two are taken without replacement.

The first counter is green. Work out the probability that the second is yellow.

1 Removing a green counter does not change the number of yellows, which is still 5.

2 But the total falls from 9 to 8.

3 So the probability is $\frac{5}{8}$.

Answer: $\frac{5}{8}$ (0.625)

7

[3 marks]

In a group of 50 people, 30 are male. 12 of the males and 8 of the females wear glasses.

(a) A male is chosen at random. Work out the probability that he wears glasses.

(b) A female is chosen at random. Work out the probability that she wears glasses.

1 (a) Restrict to the 30 males, of whom 12 wear glasses.

2 $\frac{12}{30} = \frac{2}{5}$

3 (b) There are $50 - 30 = 20$ females, of whom 8 wear glasses.

4 $\frac{8}{20} = \frac{2}{5}$

5 The two answers happen to be equal here, which suggests glasses-wearing is not linked to sex in this group.

Answer: (a) $\frac{2}{5}$ (b) $\frac{2}{5}$

8

[3 marks]

Two cards are dealt from a shuffled pack of 52 playing cards, without replacement.

Work out the probability that both are hearts. (There are 13 hearts in a pack.)

1 $P(\text{first heart}) = \frac{13}{52} = \frac{1}{4}$

2 One heart has gone, so 12 hearts remain out of 51 cards: $\frac{12}{51} = \frac{4}{17}$

3 Multiply: $\frac{1}{4} \times \frac{4}{17}$

4 $= \frac{1}{17}$

Answer: $\frac{1}{17}$ (about 0.0588)

9

[4 marks]

80 students were asked how they travel to school. The table is incomplete.

	Bus	Walk	Total
Year 10	27		45
Year 11			
Total	50		80

A student who takes the bus is chosen at random. Work out the probability that the student is in Year 11.

1 Complete the table. Year 10 who walk: $45 - 27 = 18$

2 Year 11 total: $80 - 45 = 35$

3 Year 11 on the bus: $50 - 27 = 23$ (and so $35 - 23 = 12$ walk).

4 Check the walk column: $18 + 12 = 30$, and $50 + 30 = 80$ ✓

5 The condition restricts us to the 50 bus users, of whom 23 are in Year 11.

Answer: $\frac{23}{50}$ (0.46)

10

[4 marks]

A bag contains 3 red, 4 blue and 5 green counters. Two counters are taken at random without replacement.

(a) Work out the probability that both are red.

(b) The first counter is blue. Work out the probability that the second is green.

1 There are $3 + 4 + 5 = 12$ counters altogether.

2 (a) $\frac{3}{12} \times \frac{2}{11} = \frac{6}{132} = \frac{1}{22}$

3 (b) Removing a blue counter leaves all 5 greens in place.

4 The total is now 11, so the probability is $\frac{5}{11}$.

Answer: (a) $\frac{1}{22}$ (b) $\frac{5}{11}$

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

$P(A) = 0.4$ and $P(A \text{ and } B) = 0.1$

Work out $P(B | A)$.

1 $P(B | A) = \frac{P(A \text{ and } B)}{P(A)}$

2 $= \frac{0.1}{0.4} = 0.25$

Answer: 0.25

2

[2 marks]

A bag contains 7 red and 5 blue counters. Two are taken without replacement, and the first is red.

Write down the probability that the second is also red.

1 One red counter has been removed, so 6 reds remain.

2 The total has also dropped by one, from 12 counters to 11.

Answer: $\frac{6}{11}$

3

[3 marks]

A bag contains 6 red and 6 blue counters. Two are taken without replacement.

Work out the probability that both are red.

$$1 \quad P(\text{first red}) = \frac{6}{12} = \frac{1}{2}$$

$$2 \quad P(\text{second red}) = \frac{5}{11}$$

$$3 \quad \frac{6}{12} \times \frac{5}{11} = \frac{30}{132}$$

$$4 \quad = \frac{5}{22}$$

Answer: $\frac{5}{22}$ (about 0.227)

4

[3 marks]

Using the same bag (6 red, 6 blue, two taken without replacement),

work out the probability that exactly one counter is red.

$$1 \quad \text{Red then blue: } \frac{6}{12} \times \frac{6}{11} = \frac{36}{132}$$

$$2 \quad \text{Blue then red: } \frac{6}{12} \times \frac{6}{11} = \frac{36}{132}$$

$$3 \quad \text{Add: } \frac{72}{132}$$

$$4 \quad \text{Divide top and bottom by 12: } \frac{6}{11}$$

Answer: $\frac{6}{11}$ (about 0.545)

5

[4 marks]

In a survey of 60 people, 35 own a dog, 28 own a cat and 12 own both.

(a) A cat owner is chosen at random. Work out the probability they also own a dog.

(b) A dog owner is chosen at random. Work out the probability they also own a cat.

(c) Work out the probability that a randomly chosen person owns neither.

$$1 \quad \text{Fill in a Venn diagram: both} = 12, \text{ dog only} = 35 - 12 = 23, \text{ cat only} = 28 - 12 = 16$$

$$2 \quad \text{Neither} = 60 - (23 + 12 + 16) = 60 - 51 = 9$$

$$3 \quad \text{(a) Restrict to the 28 cat owners: } \frac{12}{28} = \frac{3}{7}$$

$$4 \quad \text{(b) Restrict to the 35 dog owners: } \frac{12}{35}$$

$$5 \quad \text{(c) } \frac{9}{60} = \frac{3}{20} = 0.15$$

Answer: (a) $\frac{3}{7}$ (b) $\frac{12}{35}$ (c) 0.15

6

[4 marks]

5% of a population has a certain condition.

A test gives a positive result for 90% of people who have the condition, and also gives a positive result for 8% of people who do not have it.

(a) Work out the probability that a randomly chosen person tests positive.

(b) A person tests positive. Work out the probability that they actually have the condition, to 3 significant figures.

1 (a) Has condition and positive: $0.05 \times 0.9 = 0.045$

2 No condition and positive: $0.95 \times 0.08 = 0.076$

3 $P(\text{positive}) = 0.045 + 0.076 = 0.121$

4 (b) Restrict to the people who test positive.

5 $P(\text{condition} \mid \text{positive}) = \frac{0.045}{0.121} = 0.37190\dots$

6 So only about 37% of positive results are genuine, because the condition is rare.

Answer: (a) 0.121 (b) 0.372 (3 s.f.)

7

[4 marks]

A bag contains 5 red and 3 blue counters. Three are taken without replacement.

Work out the probability that all three are red.

1 $P(\text{first red}) = \frac{5}{8}$

2 $P(\text{second red}) = \frac{4}{7}$ and $P(\text{third red}) = \frac{3}{6}$

3 Multiply: $\frac{5}{8} \times \frac{4}{7} \times \frac{3}{6}$

4 $= \frac{60}{336}$

5 Divide top and bottom by 12: $\frac{5}{28}$

Answer: $\frac{5}{28}$ (about 0.179)

8

[4 marks]

$P(A) = 0.6$, $P(B \mid A) = 0.3$ and $P(B \mid \text{not } A) = 0.5$

(a) Work out $P(B)$.

(b) Work out $P(A \mid B)$.

1 (a) A and B: $0.6 \times 0.3 = 0.18$

2 Not A and B: $0.4 \times 0.5 = 0.2$

3 $P(B) = 0.18 + 0.2 = 0.38$

4 (b) $P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)} = \frac{0.18}{0.38}$

5 $= \frac{18}{38} = \frac{9}{19} \approx 0.474$

Answer: (a) 0.38 (b) $\frac{9}{19}$ (about 0.474)

9

[4 marks]

A bag contains 5 white counters and n black counters.

Two counters are taken without replacement. The probability that both are white is $\frac{5}{14}$

Work out the value of n .

$$1 \quad \text{There are } n + 5 \text{ counters, so } P(\text{both white}) = \frac{5}{n+5} \times \frac{4}{n+4}$$

$$2 \quad \frac{20}{(n+5)(n+4)} = \frac{5}{14}$$

$$3 \quad \text{Cross-multiply: } 20 \times 14 = 5(n+5)(n+4), \text{ so } 280 = 5(n^2 + 9n + 20)$$

$$4 \quad \text{Divide by 5: } 56 = n^2 + 9n + 20, \text{ so } n^2 + 9n - 36 = 0$$

$$5 \quad (n+12)(n-3) = 0, \text{ so } n = 3 \text{ (rejecting the negative solution).}$$

$$6 \quad \text{Check: } \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14} \quad \checkmark$$

Answer: $n = 3$

10

[5 marks]

A box contains 4 milk chocolates and 6 dark chocolates. Two are taken at random without replacement.

(a) Work out the probability that both are milk.

(b) Work out the probability that the second chocolate is dark.

(c) Given that the second chocolate is dark, work out the probability that the first was milk.

$$1 \quad \text{(a)} \quad \frac{4}{10} \times \frac{3}{9} = \frac{12}{90} = \frac{2}{15}$$

2 (b) There are two routes to a dark second chocolate.

$$3 \quad \text{Milk then dark: } \frac{4}{10} \times \frac{6}{9} = \frac{24}{90}$$

$$4 \quad \text{Dark then dark: } \frac{6}{10} \times \frac{5}{9} = \frac{30}{90}$$

$$5 \quad P(\text{second dark}) = \frac{54}{90} = \frac{3}{5} \text{ (the same as } P(\text{first dark}), \text{ which is no accident).}$$

6 (c) Restrict to the cases where the second is dark.

$$7 \quad P(\text{first milk} \mid \text{second dark}) = \frac{24}{90} \div \frac{54}{90} = \frac{24}{54} = \frac{4}{9}$$

Answer: (a) $\frac{2}{15}$ (b) $\frac{3}{5}$ (c) $\frac{4}{9}$